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(54) **SYSTEMS AND METHODS FOR
CONVERSION OF A BIOMASS INTO
BIOFUEL USING A GEOTHERMAL HEAT
SOURCE**

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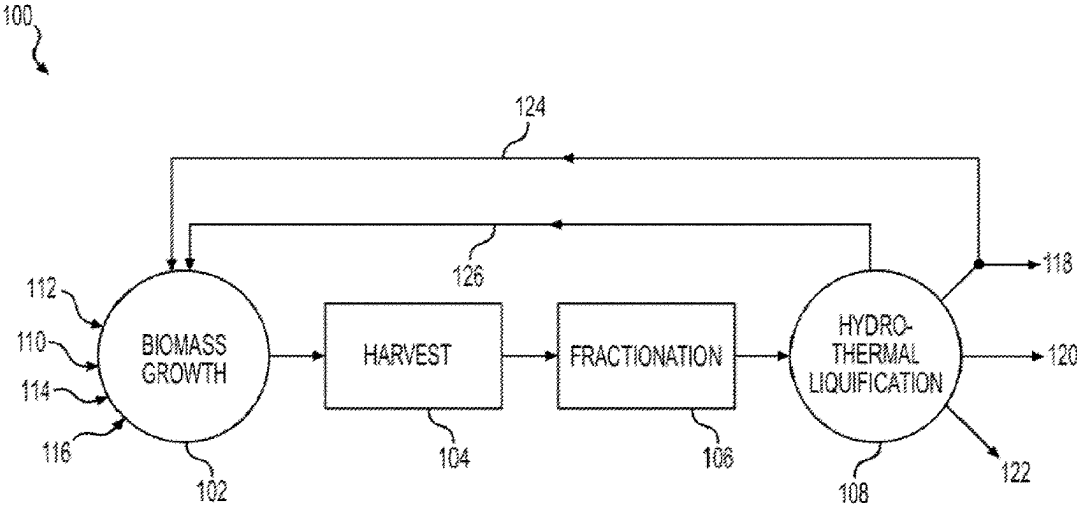
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(57) **ABSTRACT**
A system for converting a biomass into a biofuel including
a biomass processing station arranged to receive the biomass
from a biomass harvester, output the biomass to a hydro-
thermal liquefaction (HTL) converter, and receive a pro-
cessed biomass from the HTL converter. The system
includes a conduit arranged to transport the biomass from
the biomass processing station to the HTL converter and
transport the processed biomass from the HTL converter to
the biomass processing station. The HTL converter includes
a heat exchanger arranged to transfer thermal energy from a
geothermal heat source to the biomass to convert the bio-
mass into the processed biomass. The system also includes
a controller arranged to monitor conditions of the biomass at
locations along the conduit and adjust operations of com-
ponents along the conduit to, thereby, adjust the conditions
of the biomass at one or more locations along the conduit.

17 Claims, 7 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/570,068, filed on
Jan. 6, 2022, now Pat. No. 11,505,772.

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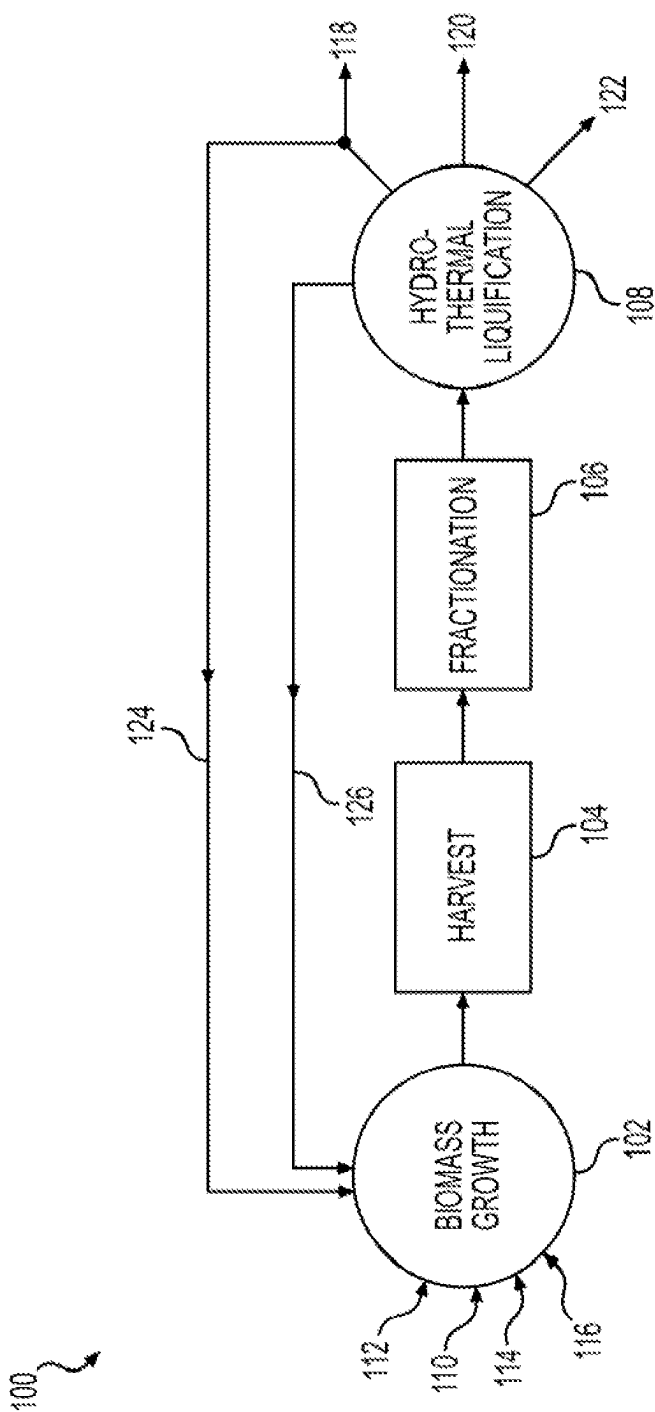
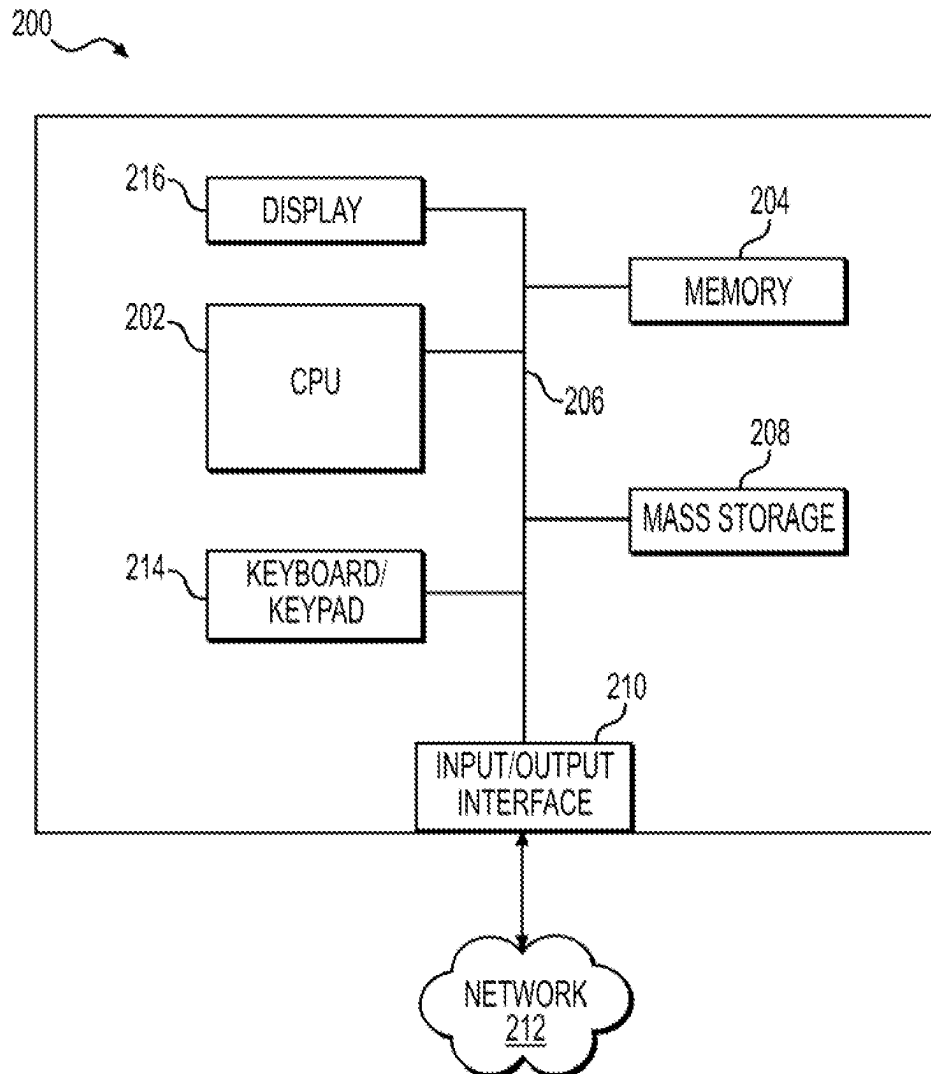


FIG. 1

**FIG. 2**

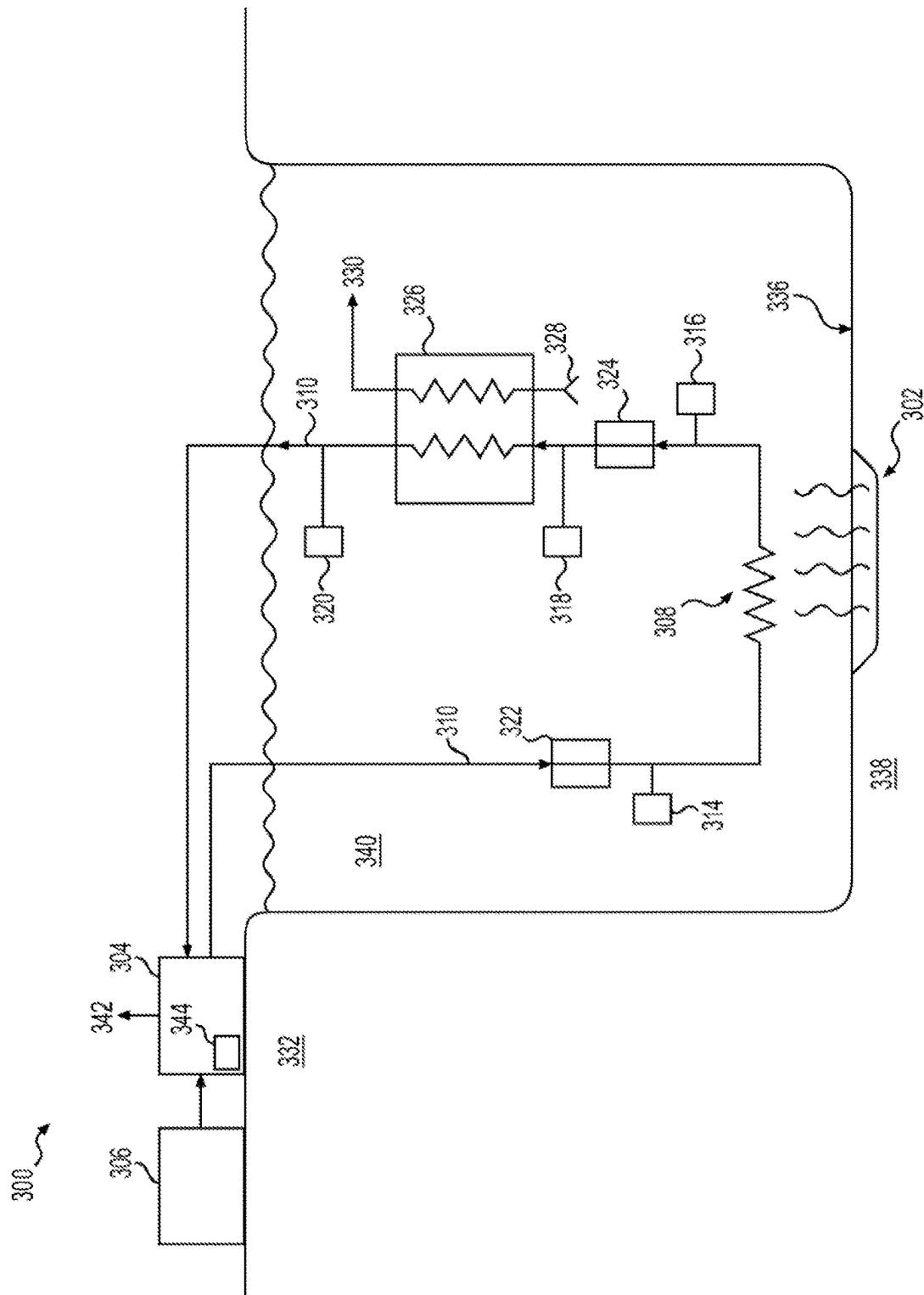
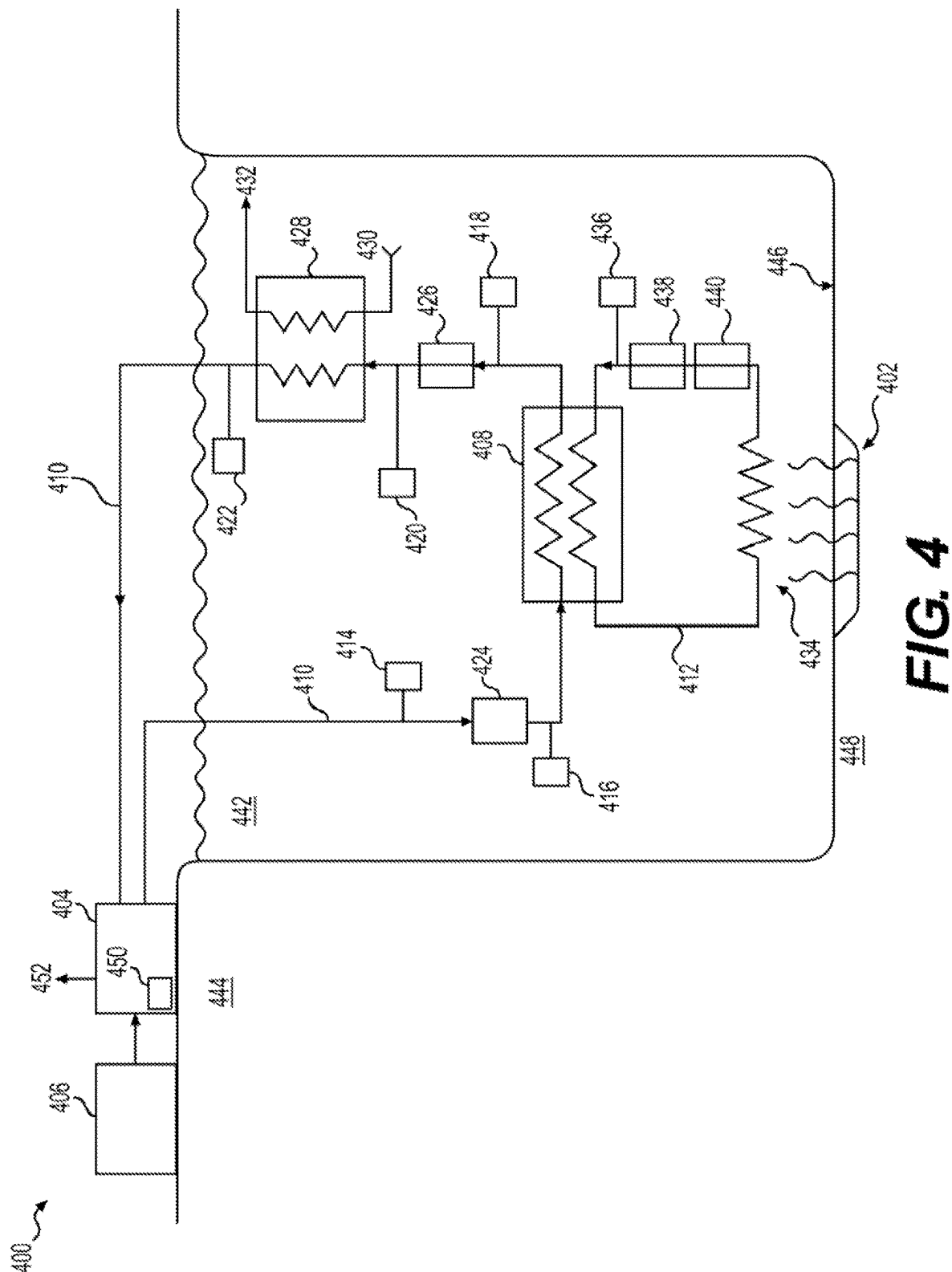
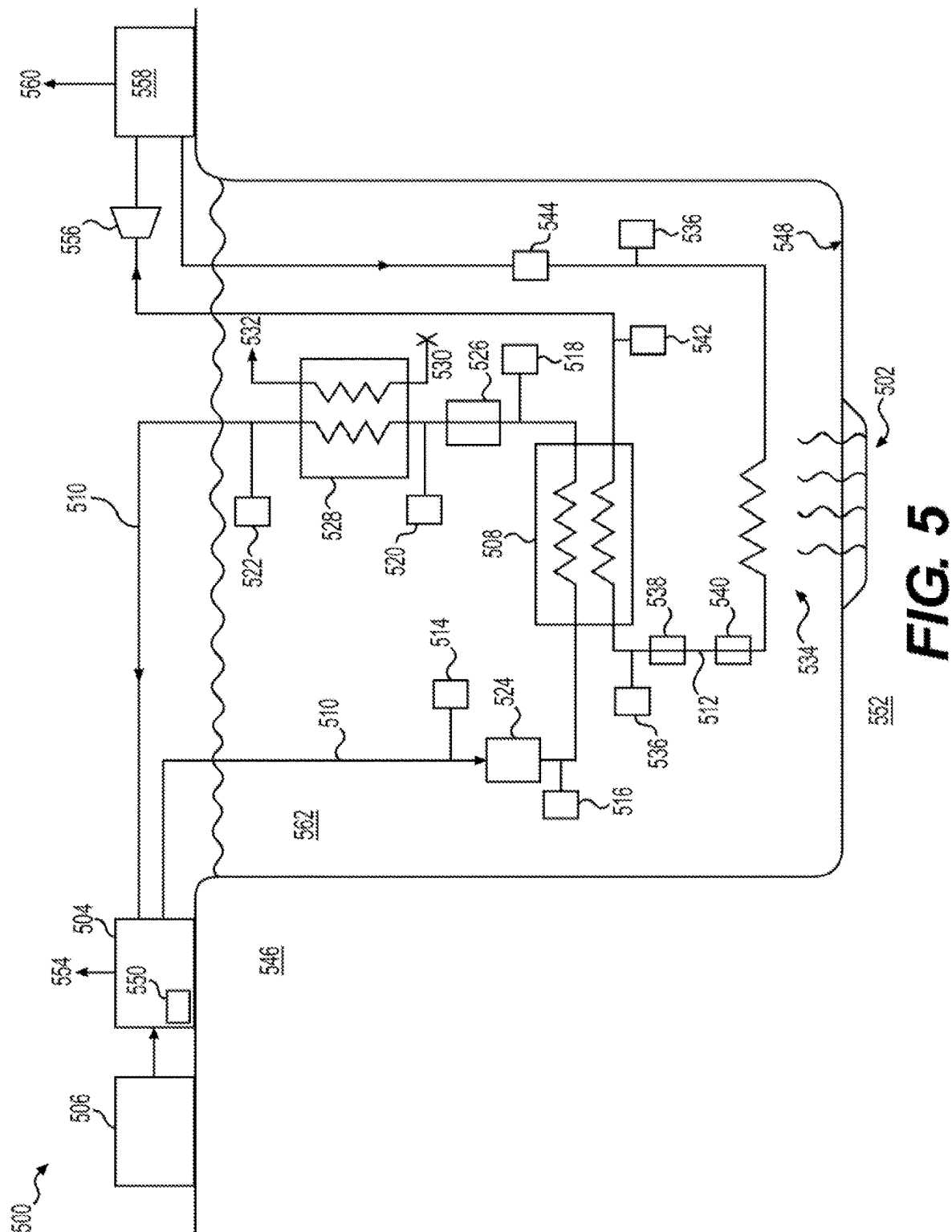
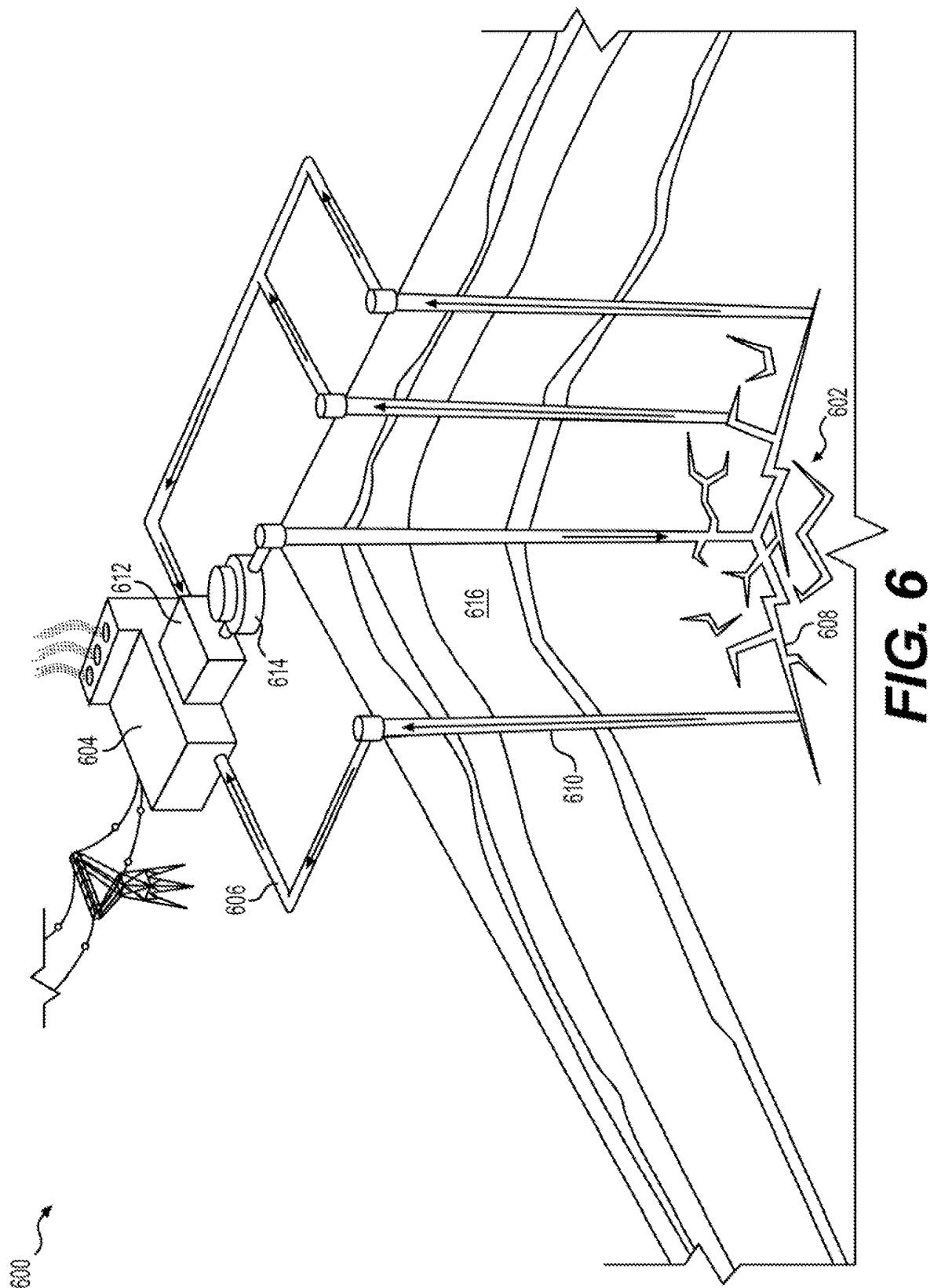
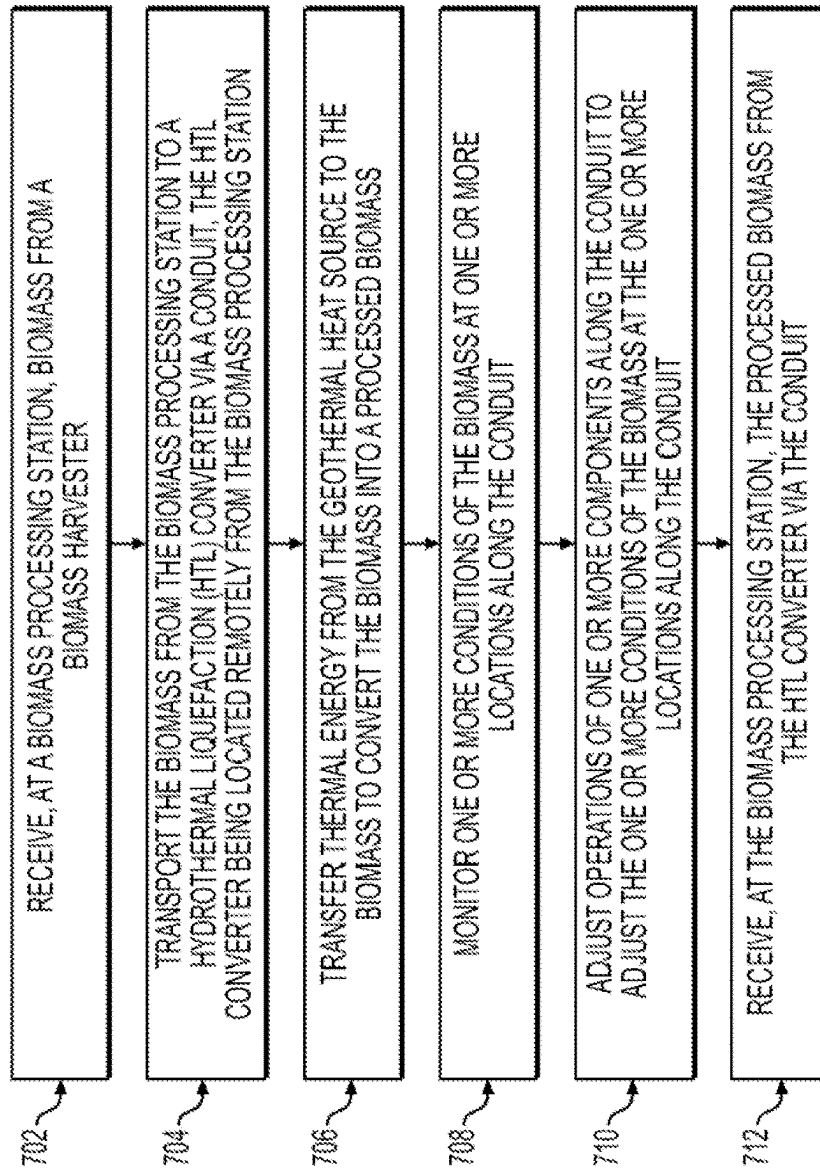


FIG. 3







**FIG. 7**

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SYSTEMS AND METHODS FOR CONVERSION OF A BIOMASS INTO BIOFUEL USING A GEOTHERMAL HEAT SOURCE

REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/990,032, filed on Nov. 18, 2022, which issued as U.S. Pat. No. 11,959,055, which is a continuation of U.S. patent application Ser. No. 17/570,068, filed on Jan. 6, 2022, which issued as U.S. Pat. No. 11,505,772, the entire contents of the which are incorporated herein by reference.

TECHNICAL FIELD

This application relates generally to converting a biomass into a biofuel and, more particularly, to geothermal biofuel production based on algae or seaweed.

BACKGROUND

Fuel derived from algae, often referred to as the ‘third-generation biofuel’, provides advantages over earlier biofuels based on plant crops including sugar cane and corn (e.g., the “first generation of biofuel”) and vegetable or animal waste streams (e.g., the “second generation of biofuel”).

Algae biofuels have advantages that include higher biofuel yields compared to previous systems, a diverse set of possible fuel types including biodiesel, butanol, ethanol and jet fuel, as well as the capability to deploy large-scale algae cultivation in open ponds or more advanced closed-loop systems. Biofuel production from algae can be done on land unsuitable for food crops, removing concerns that algae biofuel feedstock crops would compete with food producers.

Algae-based biofuel companies have unsuccessfully attempted to chemically engineer algae capable of scaling up to produce tens of millions of gallons of fuel at prices competitive with fossil fuels. Fuel conversion from algae has been generally based on the feedstock’s high concentrations of lipids, i.e., fatty, oil-containing acid molecules that can be extracted to create biofuels. Algae can be converted into various types of fuels, depending on the technique and the part of the cells used. The lipid, or oily part of the algae biomass can be extracted and converted into biodiesel through a process similar to that used for any other vegetable oil or converted in a refinery as a replacement for petroleum-based fuels. Alternatively, or following lipid extraction, the carbohydrate content of algae can be fermented into bioethanol or butanol fuel.

Biofuel is typically produced from algae via dehydration or hydrothermal liquefaction. Hydrothermal liquefaction implements a continuous process that subjects harvested wet algae to high temperatures and pressures, e.g., 350° C. (662° F.) and 3,000 pounds per square inch (21,000 kPa). Unfortunately, the cost of generating such high temperatures and pressures has traditionally exceeded the value of the resulting algae-based biofuel product. Such products include crude oil, which can be further refined into aviation fuel, gasoline, or diesel fuel using one or many upgrading processes. Other outputs from the conversion process can include clean water, gas fuel or nutrients such as nitrogen, phosphorus, and potassium.

Despite significant efforts to develop cost-efficient and scalable conversion processes, practical biofuel production systems have yet to be achieved. Technical problems remain in relation to the energy balance of lipid extraction, main-

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taining suitable growing conditions in open waters, power consumption, and the immense volumes of water, CO₂ and fertilizer required to allow the algae to photosynthesize fast enough in large volumes.

While there have been many research efforts to develop algae biofuel production systems, none of the existing production efforts have been capable of converting algae into biofuel in a sufficiently scalable and cost-efficient manner.

SUMMARY

The application, in various implementations, addresses deficiencies associated with converting a biomass such as algae into a biofuel.

Systems and methods described herein utilize a geothermal energy source within an ocean and/or under the Earth to enable an efficient hydrothermal liquefaction (HTL) process for thermochemical conversion of wet algae and/or microalgae to produce a liquid energy carrier referred to ‘bio oil,’ ‘biofuel,’ or ‘biocrude’, along with gaseous, aqueous and/or solid by-products.

In one aspect, a system for converting a biomass into a biofuel includes a biomass processing station arranged to: receive the biomass from a biomass harvester, output the biomass to a hydrothermal liquefaction (HTL) converter, and receive a processed biomass from the HTL converter. The system includes a conduit arranged to transport the biomass from the biomass processing station to the HTL converter and transport the processed biomass from the HTL converter to the biomass processing station. The HTL converter includes a heat exchanger arranged to transfer thermal energy from a geothermal heat source to the biomass to convert the biomass into the processed biomass. The system further includes a controller arranged to monitor one or more conditions of the biomass at the one or more locations along the conduit and adjust operations of one or more components along the conduit to, thereby, adjust the one or more conditions of the biomass at the one or more locations along the conduit.

In one implementation, the system includes one or more sensors located at one or more sections of the conduit that enable the controller to monitor the one or more conditions of the biomass by receiving sensor data associated with one or more conditions of the biomass from the one or more sensors. The biomass may include an algae and the processed biomass may include bio oil, biocrude, and/or biofuel. In certain configurations, converting the biomass into a processed biomass includes depolymerization of the biomass to bio oil. The controller may be arranged to adjust the operations of the one or more components to optimize the one or more conditions of the biomass to optimize a quality and/or yield of the processed biomass. The one or more components may include at least one of a pump and a flow control valve. The one or more conditions of the biomass may include temperature, rate of temperature change, pressure, flow rate, catalyst concentration, and/or residence time within the HTL converter.

The geothermal heat source may include a volcano, volcanic vent, and/or a hydrothermal vent. The HTL converter may reside within a body of water and a substantial portion of the conduit may be surrounded by the body of water. The biomass processing station and biomass harvester may be in proximity with each other within the body of water.

In another aspect, a method for converting a biomass into a biofuel includes: receiving, at a biomass processing sta-

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tion, the biomass from a biomass harvester; transporting the biomass from the biomass processing station to a hydrothermal liquefaction (HTL) converter via a conduit, where the HTL converter is located remotely from the biomass processing station and within proximity to a geothermal heat source; transferring thermal energy from the geothermal heat source to the biomass to convert the biomass into a processed biomass; monitoring one or more conditions of the biomass at one or more locations along the conduit; adjusting operations of one or more components along the conduit to adjust the one or more conditions of the biomass at the one or more locations along the conduit; and receiving, at the biomass processing station, the processed biomass from the HTL converter via the conduit.

In a further aspect, a system for converting a biomass into a biofuel includes a biomass processing station arranged to: receive the biomass from a biomass harvester, output the biomass to a HTL converter, and receive a processed biomass from the HTL converter. The system includes a first conduit arranged to transport the biomass from the biomass processing station to the HTL converter and transport the processed biomass from the HTL converter to the biomass processing station. The HTL converter includes a first heat exchanger arranged to transfer thermal energy from a superheated fluid to the biomass to convert the biomass into the processed biomass. The system includes a second conduit arranged to transport the superheated fluid to the HTL converter from a second heat exchanger in proximity to a geothermal heat source. The second heat exchanger is arranged to transfer thermal energy from the geothermal heat source to the superheated fluid. The system further includes a controller arranged to perform at least one of: i) monitor one or more conditions of the biomass at the one or more locations along the first conduit and adjust operations of one or more components along the first conduit to, thereby, adjust the one or more conditions of the biomass at the one or more locations along the first conduit, and ii) monitor one or more conditions of the superheated fluid at one or more locations along the second conduit and adjust operations of one or more components along the second conduit to, thereby, adjust the one or more conditions of the superheated fluid at the one or more locations along the second conduit.

Any two or more of the features described in this specification, including in this summary section, may be combined to form implementations not specifically described in this specification. Furthermore, while this specification may refer to examples of systems, methods, and devices related algae or seaweed producing bioreactors, such techniques also apply equally to bioreactors arranged to cultivate other organisms or products.

The systems and methods described herein are capable of refining any biomass including terrestrial ones, and also plastic waste as a fraction of the feedstock. In some implementations, the primary outputs of the systems and methods include Biocrude, biofuels and hydrogen. In the case of hydrogen, implementations described herein can yield a set of carbon compound byproducts that can be sequestered or have applications in, for example, concrete and other industrial materials. Hence, as a platform or process, the systems and methods described herein have the potential to be a carbon-negative solution for Bioenergy with Carbon Capture and Storage (BECCS), based on a lack of emissions as a refining system. The systems and methods described herein, in various implementations, can also, depending on what material is being processed, produce nitrogen, phosphorous, and other valuable material byproducts including,

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in some cases, rare earth metals such as Scandium, or a range of other rare earth elements (REEs). Paltry percentage, but if we process 1 M tons a day these become notable. Other applications of the systems and methods described herein may render materials for a material market, e.g., bioplastics. With a capacity to accept waste plastics along with biomass, the geothermal and/or hydrothermal liquefaction systems and methods described herein can also serve as a recycling solution. A geothermal HTL (GeoHTL) refinery such as described with respect to the systems and methods herein can also be deployed on land, on an ocean surface, or on the ocean floor.

The details of one or more implementations are set forth in the accompanying drawings and the following description. Other features and advantages will be apparent from the description and drawings, and from the claims.

DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram of an exemplary biomass to biofuel conversion process;

FIG. 2 shows a diagram of a computer system for a controller arranged to control operations of a biomass to biofuel conversion system;

FIG. 3 shows a system for converting a biomass into biofuel using a geothermal heat source;

FIG. 4 shows another system for converting a biomass into biofuel using a geothermal heat source;

FIG. 5 shows a further system for converting a biomass into biofuel using a geothermal heat source;

FIG. 6 shows a land-based system for converting a biomass into biofuel using a geothermal heat source; and

FIG. 7 shows a process for converting a biomass into biofuel using a geothermal heat source.

Like reference numerals in different figures indicate like elements.

DETAILED DESCRIPTION

The application, in various implementations, addresses deficiencies associated with converting a biomass, such as algae, into a biofuel. This application describes exemplary systems, methods, and devices that effectively and efficiently implement biomass conversion to biofuel by using a geothermal heat source to enable depolymerization of the biomass to bio oil. The conversion process may be enhanced by controlling conditions associated with the biomass during conversion including controlling temperature, rate of temperature change, pressure, catalyst concentration, and/or flow rate of the biomass through an HTL converter to, thereby, optimize production yield of the bio oil and/or biofuel derived from the conversion. The optimization may be further enhanced by monitoring conditions of the biomass and/or bio oil along a conduit transporting the materials to and from the HTL converter using sensors to provide sensor data to a controller that uses AI and/or ML to process the sensor data and dynamically adjust operations of various conversion system components to adjust one or more conditions within the conduit and/or HTL converter, which optimizes biofuel quality and/or yield, or optimizes biofuel characteristics for a particular use.

FIG. 1 is a diagram of an exemplary biomass to biofuel conversion process **100**. The conversion process **100** includes a biomass growth stage **102**, harvesting stage **104**, fracturing stage **106**, and a hydrothermal liquefaction (HTL) stage **108**. During biomass growth stage **102**, a carbon source (e.g., algae seed) **110** is combined at least with light

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112, water **114**, and nutrients **116** to facilitate growth of a biomass (e.g., algae) within, for example, a photobioreactor. Detailed descriptions of on-shore, closed loop, open, and deepwater photobioreactors are provided in U.S. Provisional Patent Application No. 63/172,407, entitled “Photobioreactor Systems and Methods,” U.S. patent application Ser. No. 17/564,779, filed on Dec. 29, 2021, entitled “Photobioreactor Systems and Methods,” and U.S. patent application Ser. No. 17/564,814, filed on Dec. 29, 2021, entitled “Systems and Methods for Deepwater Photobioreactor,” the entire contents of which are incorporated herein by reference.

Once the biomass is formed during the growth stage **102**, the biomass is harvested from, for example, a photobioreactor. During harvesting, the harvested biomass may be further fractioned via a fractionation stage **106** to reduce the size of the biomass materials. The biomass is then depolymerized during HTL stage **108** into bio oil or biofuel. By products of the HTL stage **108** may include CO₂ rich gas that can be returned to the biomass growth stage **102** via feedback **124** to enhance the biomass growth process. Other materials such as water and nutrients may be recycled back via a feedback **126** to biomass growth stage **102** for more efficient biofuel production. Solid residue **120** including, for example, char may be extracted from the biofuel during stage **108**. Bio crude **122**, biofuel, and/or bio oil are also extracted for storage, use, or further processing as a fuel.

FIG. 2 includes a block diagram of a computer system **200** for performing the functions of a computer such as for the controller of FIGS. 3, 4, 5, and 6. The exemplary computer system **200** includes a central processing unit (CPU) **202**, a memory **204**, and an interconnect bus **206**. The CPU **202** may include a single microprocessor or a plurality of microprocessors for configuring computer system **200** as a multiprocessor system. The memory **204** illustratively includes a main memory and a read only memory. The computer **200** also includes the mass storage device **208** having, for example, various disk drives, tape drives, etc. The main memory **204** also includes dynamic random access memory (DRAM) and high-speed cache memory. In operation, the main memory **204** stores at least portions of instructions and data for execution by the CPU **202**.

The mass storage **208** may include one or more magnetic disk or tape drives or optical disk drives or solid state memory, for storing data and instructions for use by the CPU **202**. At least one component of the mass storage system **208**, preferably in the form of a disk drive, solid state, or tape drive, stores the database used for processing sensor data from sensor array **116** and running AI and/or ML engines and/or neural networks for controlling biofuel conversion systems **300**, **400**, and **500**. The AI and/or ML engines may implement ANNs and/or Deep-learning architectures such as deep neural networks, deep belief networks, recurrent neural networks and convolutional neural networks to dynamically adjust environmental conditions within the conduits, HTL converter, or other components of systems **300**, **400**, **500**. To effect automated control of system **300**, **400**, **500**, and **600**, computer **200** may send control signals to various components **322**, **324**, **326**, **424**, **426**, **428**, **524**, **526**, and **528** of systems **300**, **400**, **500**, or components of system **600**, to either, open, close, turn on, turn off, adjust flow rate, adjust mixing rate, and cooling rate, to optimize bio crude and/or biofuel production within the HTL converter of system **300**, **400**, **500**, or **600**. The mass storage system **208** may also include one or more drives for various portable media, such as a floppy disk, flash drive, a compact disc read only memory (CD-ROM, DVD, CD-RW, and variants), memory stick, or an integrated circuit non-volatile

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memory adapter (i.e. PC-MCIA adapter) to input and output data and code to and from the computer system **200**. In some implementations, computer **200** and/or controller **118** may control multiple bioreactors concurrently via a data network such as network **212**. Controller **118** may coordinate operations among the multiple bioreactors to optimize output production and/or yield among the multiple bioreactors. Network **212** may include a wireless, Adhoc, and/or mobile network, supporting multiple computing server implementations in a cloud computing environment. Various environmental sensors and/or multiple biofuel conversion systems, harvesters (e.g., photobioreactors), and biomass processing stations, and/or robots servicing component of these systems may be communicatively connected via network **212** as, for example, Internet-of-Things (IoT) capable systems and/or devices. In some implementations, network **212** may enable computer **200** and/or controller **118** to coordinate operations of multiple photobioreactors by using predictive analytics to process, for example, global position system (GPS) data and other big data to coordinate operations and control of multiple concurrently operating bioreactors over a geographic area. In certain implementations, network **212** may enable collections of, for example, GPS data from multiple bioreactors and use an ML program to enhance security and/or performance for biofuel production on land or in the sea.

The computer system **200** may also include one or more input/output interfaces for communications, shown by way of example, as interface **210** and/or transceiver for data communications via the network **212**. The data interface **210** may be a modem, an Ethernet card or any other suitable data communications device. To provide the functions of a computer **102**, the data interface **210** may provide a relatively high-speed link to a network **212**, such as an intranet, or the Internet, either directly or through another external interface. The communication link to the network **212** may be, for example, optical, wired, or wireless (e.g., via satellite or cellular network). Alternatively, the computer system **200** may include a mainframe or other type of host computer system capable of Web-based communications via the network **212**. The computer system **200** may include software for operating a network application such as a web server and/or web client.

The computer system **200** may also include suitable input/output ports, that may interface with a portable data storage device, or use the interconnect bus **206** for interconnection with a local display **216** and keyboard **214** or the like serving as a local user interface for programming and/or data retrieval purposes. The display **216** and/or display **120** may include a touch screen capability to enable users to interface with the system **200** by touching portions of the surface of the display **216**. Remote operations personnel may interact with the system **200** for controlling and/or programming the system from remote terminal devices via the network **212**.

The computer system **200** may run a variety of application programs and store associated data in a database of mass storage system **208**. One or more such applications may include a biofuel conversion controller **344**, **450**, **550**, and **612** that controls various components of system **300**, **400**, **500**, and **600** during the biofuel conversion process.

The components contained in the computer system **200** may enable the computer system to be used as a server, workstation, personal computer, network terminal, mobile computing device, and the like. As discussed above, the computer system **200** may include one or more applications that enable cleaning and sanitization of a footwear sole or

soles. The system **200** may include software and/or hardware that implements a web server application. The web server application may include software such as HTML, XML, WML, SGML, PHP (Hypertext Preprocessor), CGI, and like languages.

The foregoing features of the disclosure may be realized as a software component operating in the system **200** where the system **200** includes UNIX workstation, a Windows workstation, a LINUX workstation, or other type of workstation. Other operating systems may be employed such as, without limitation, Windows, MAC OS, and LINUX. In some aspects, the software can optionally be implemented as a C language computer program, or a computer program written in any high level language including, without limitation, JavaScript, Java, CSS, Python, PHP, Ruby, C++, C, Shell, C#, Objective-C, Go, R, TeX, VimL, Perl, Scala, CoffeeScript, Emacs Lisp, Swift, Fortran, or Visual BASIC. Certain script-based programs may be employed such as XML, WML, PHP, and so on. The system **200** may use a digital signal processor (DSP).

As stated previously, the mass storage **208** may include a database. The database may be any suitable database system, including the commercially available Microsoft Access database, and can be a local or distributed database system. A database system may implement Sybase and/or an SQL Server. The database may be supported by any suitable persistent data memory, such as a hard disk drive, RAID system, tape drive system, floppy diskette, or any other suitable system. The system **200** may include a database that is integrated with the system **200**, however, it is understood that, in other implementations, the database and mass storage **208** can be an external element.

In certain implementations, the system **200** may include an Internet browser program and/or to be configured to operate as a web server. In some configurations, the client and/or web server may be configured to recognize and interpret various network protocols that may be used by a client or server program. Commonly used protocols include Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Telnet, and Secure Sockets Layer (SSL), and Transport Layer Security (TLS), for example. However, new protocols and revisions of existing protocols may be frequently introduced. Thus, to support a new or revised protocol, a new revision of the server and/or client application may be continuously developed and released.

The computer system **200** may include a web server running a Web 2.0 application or the like. Web applications running on system **200** may use server-side dynamic content generation mechanisms such, without limitation, Java servlets, CGI, PHP, or ASP. In certain embodiments, mashed content may be generated by a web browser running, for example, client-side scripting including, without limitation, JavaScript and/or applets on a wireless device.

In certain implementations, system **200** and/or controller **118** may include applications that employ asynchronous JavaScript+XML (Ajax) and like technologies that use asynchronous loading and content presentation techniques. These techniques may include, without limitation, XHTML and CSS for style presentation, document object model (DOM) API exposed by a web browser, asynchronous data exchange of XML data, and web browser side scripting, e.g., JavaScript. Certain web-based applications and services may utilize web protocols including, without limitation, the services-orientated access protocol (SOAP) and representational state transfer (REST). REST may utilize HTTP with XML.

The system **200** may also provide enhanced security and data encryption. Enhanced security may include access control, biometric authentication, cryptographic authentication, message integrity checking, encryption, digital rights management services, and/or other like security services. The security may include protocols such as IPSEC and IKE. The encryption may include, without limitation, DES, 3DES, AES, RSA, and any like public key or private key based schemes.

FIG. 3 shows a system **300** for converting a biomass into biofuel using a geothermal heat source **302**. System **300** includes a biomass processing station **304** arranged to receive the biomass from a biomass harvester **306** and then output the biomass to a HTL converter **308**. The processing station **304** also receives a processed biomass from the HTL converter **308**. The system **300** uses a conduit **310** to transport the biomass from the biomass processing station **304** to the HTL converter **308** and transport the processed biomass from the HTL converter **308** to the biomass processing station **304**. HTL converter **308** includes a heat exchanger arranged to transfer thermal energy from geothermal heat source **302** to the biomass being transported through conduit **310** to convert the biomass into the processed biomass. System **300** also includes a controller **344** that is arranged to monitor one or more conditions of the biomass via sensors **314**, **316**, **318**, and **320** at the one or more locations along the conduit **310** and adjust operations of one or more components **322**, **324**, and **326** along the conduit **310** to, thereby, adjust the one or more conditions of the biomass at the one or more locations along the conduit **310**.

The conditions monitored by sensors **314-320** may include, without limitation, temperature, rate of temperature change, pressure, flow rate, catalyst concentration, and/or residence time the biomass within HTL converter **308**. Controller **344** may monitor the one or more conditions of the biomass by receiving sensor data associated with one or more conditions of the biomass from one or more of sensors **314-320**. The biomass may include an algae while the processed biomass may include bio oil, bio crude, and/or biofuel. The conversion of the biomass to biofuel may include depolymerization of the biomass to bio oil.

In various implementations, controller **344** is configured to adjust the operations of one or more of components **322**, **324**, and **326** to optimize conditions of the biomass and, thereby, optimize a quality and/or yield of the processed biomass. Components **322** and **324** may include flow regulators such as, for example, a pump and/or a flow control valve. A pump may be a variable speed pump to adjust flow rate within conduit **310**. A flow control valve may be adjustable to adjust flow rate with conduit **310**. Heat exchanger **326** may be controlled by controller **344** to regulate an output temperature of from heat exchanger **326** before the processed biomass, e.g., biofuel, is returned to processing station **304**. Heat exchanger **326** may receive cooling water at a temperature set by the depth of water inlet **328** and pass water through to extract heat from the processed biomass also passing through heat exchanger **326** via conduit **310**. The flow rate of cooling water through heat exchanger **326** may be regulated by a flow regulator controlled by controller **344**. The heated cooling water is then expelled into the surround body of water **340**, e.g., the ocean. In this way, the temperature of the processed biomass is reduced before handling at processing station **304**. Although not shown in FIG. 3, controller **344** is in electrical communications, e.g., via electrical conduits, with sensors

314-320 and components **322-326** to enable electrical communications associated with controlling conditions of the biomass in system **300**.

The geothermal heat source **302** may include, without limitation, a volcano, volcanic vent, or a hydrothermal vent. Although HTL converter **308** is shown adjacent to geothermal heat source **302**, HTL converter **308** may extend through source **302** within the earth **338** below the ocean floor **336**. HTL converter **308** may include a section of conduit **310** that extends through and/or near source **302**. The location of HTL converter **308** with respect to source **302** may depend on the desired amount of heat exchange and temperature needed to convert the biomass into a processed biomass. HTL converter **308** may include a single pipe (i.e., section of conduit **310**) that extends through a region of the geothermal heat source **302**. The pipe may be substantial straight, U-shaped, serpentine, and/or curved in portions. HTL converter **308** may include multiple parallel pipes and/or capillaries with a common inlet and outlet that interface with conduit **310**.

As illustrated in FIG. 3, HTL converter **308** may reside within a body of water **340** such as the ocean where a substantial portion of conduit **310** is surrounded by the body of water **340**. While FIG. 3 shows the harvester **306** and biomass processing station **304** on land **322**, the biomass processing station **304** and biomass harvester **306** may reside in proximity with each other within the body of water **340**. The biomass processing station **304** may be located on a vessel, rig, or platform on the body of water **340**. The harvester **306** may include one or more photobioreactors or deepwater photobioreactors in the body of water **340** and in proximity to the system **300** and/or station **304**. The harvester **306** may be less than or equal to about 5 miles (about 8 Km), 3 miles (about 4.8 Km), 1 mile (about 1.6 Km), 0.5 miles (about 0.8 Km), 0.25 miles (about 0.4 Km) from system **300** and/or biomass processing station **304**.

In certain implementations, controller **344** is arranged to enable automated control of components **322**, **324**, and **326** of the biofuel conversion system **300**. Controller **344** may include a processor running artificial intelligence (AI) and/or machine learning (ML), neural networks, Bayesian networks, and/or fuzzy logic to process sensor data received from one or more of sensors **314-316** and control various biomass parameters of system **300** including, without limitation, biomass flow rates, temperature, rate of temperature change, nutrient concentrations, PH levels, dissolved gases concentrations, and/or residence time within HTL converter **308**. Controller **344** may implement Artificial Neural Networks (ANN) and/or Deep-learning architectures such as deep neural networks, deep belief networks, recurrent neural networks and convolutional neural networks to dynamically adjust biomass and/or conduit **310** conditions within system **300**, **400**, **500**, and **600**. Controller **344** may implement supervised learning, reinforcement learning, and/or unsupervised learning. Reinforced learning may include game theory, control theory, operations research, information theory, and/or simulation-based optimization to dynamically adjust biomass and/or conduit **310** conditions within system **300**, **400**, **500**, and **600**. The system **300** cultivation and/or conversion environment may be represented as a Markov decision process (MDP). Controller **344** may create multiple decision trees that solve multiple conversion optimization problems. Controller **344** may use Bayesian networks to optimize an biomass to biofuel conversion process.

Controller **344** may use one or more neural networks, such multilayer perceptrons (MLPs), convolutional neural networks (CNNs), or deep Boltzman machines (DBM) that

are trained to compute a function that maps an input vector to an output vector. The N-element output vector may convey estimates of the probabilities of N biomass conversion settings. In some implementations, controller **344** uses a recurrent neural network (RNN) where its neurons send feedback signals to each other to enable dynamic temporal behavior. Controller **344** may use an enhanced RNN referred to as long short-term memory (LSTM) and/or hierarchical temporal memory (HTM). Controller **344** may use a combination of the aforementioned AI algorithms to form a hybrid control system. A decision tree is a generic term that describes a decision process that may use one or more attributes at each node and/or use an information theoretic measure to formulate queries at each node to reach a decision on the optimal biomass conversion configuration for producing biofuel using system **300**, **400**, **500**, or **600**.

In operation in some implementations, the biomass processing station **304** receives the biomass from biomass harvester **306**. Conduit **310** transports the biomass from the biomass processing station **304** to HTL converter **308** that is located remotely from the biomass processing station **304** and within proximity to geothermal heat source **302**. One or more pumps within processing station **304**, flow regulator **322**, and or flow regulator **324** may be operated to facilitate transport of the biomass through conduit **310**. At HTL converter **308**, thermal energy is transferred from geothermal heat source **302** to the biomass, resulting in a conversion of the biomass into a processed biomass, e.g., bio crude. Controller **344** may monitor one or more conditions of the biomass at one or more locations along the conduit **310** via sensors **314-320** and, in response to sensor data received from one or more of sensors **314-320**, adjust operations of one or more components **322**, **324**, and **326** along the conduit **310** to adjust the one or more conditions of the biomass at the one or more locations along the conduit **310**. Then, the processed biomass from the HTL converter **308** is received at the biomass processing station **304** via conduit **310**. The processed biomass may then be outputted from the biomass processing station via outlet **342** to a storage facility or further processing facility. Biomass processing station may include or be part of a fuel refinery.

Hydrothermal liquefaction (HTL) is a thermal depolymerization process that is used to convert a wet biomass, and other macromolecules, into crude oil under moderate to high temperature and high pressure. The crude oil may have a high energy density with a lower heating value of, for example, 33.8-36.9 MJ/kg and 5-20 weight (wt) % of oxygen and renewable chemicals. The reaction may include homogeneous and/or heterogeneous catalysts to enhance product quality and yields. The carbon and hydrogen of an organic material, such as the biomass, can be thermodynamically converted into hydrophobic compounds with low viscosity and high solubility. Depending on the processing conditions, the biofuel can be used for heavy engines, including marine and rail or upgraded to transportation fuels, such as diesel, gasoline or jet-fuels.

In a hydrothermal liquefaction process, long carbon chain molecules in a biomass are thermally cracked and oxygen is removed in the form of H₂O (dehydration) and CO₂ (decarboxylation). These reactions result in the production of a high H/C ratio bio-oil where H/C=(wt % hydrogen/atomic wt hydrogen)/(wt % carbon/atomic wt carbon). Typical implementations of hydrothermal liquefaction operate at temperatures between 250-550° C. and high pressures of 5-25 MPa while subjected to catalysts for 20 to 60 minutes. But, in certain implementations, controller **344** may regulate higher or lower temperatures within HTL converter **308**

and/or conduit **310** to optimize gas or liquid yields of the processed biomass. At these temperatures and pressures, the water present in the biomass may become either subcritical or supercritical, depending on the conditions, and may act as a solvent, reactant, and catalyst to facilitate the reaction of the biomass to a processed biomass, e.g., a bio oil.

Controller **344**, **450**, **550**, and **612** may control the conversion parameters of the biomass to bio oil based on controlling one or more of various conditions including: feedstock composition, temperature and heating rate, pressure, flow rate, solvent composition, residence time in the HTL converter **308**, and the composition of catalysts. In some configurations, controller **344** may control residence time in HTL converter **308** by controlling the biomass flow rate through HTL converter **308**. For example, if the conduit **310** length of HTL converter **308** is 200 meters, the controller **344** may set operations of, for example, flow regulator **322** to set the biomass flow rate through HTL converter **308** to 10 m/min. This would result in biomass residence time within HTL converter of 20 minutes which is equal to the distance of 200 meters divided by the flow rate (10 m/min). Residence times of the biomass within the HTL converter **308** may be less than or equal to about 60 minutes, 40 minutes, 20 minutes, 10 minutes, 5 minutes, and 1 minute. Controller **344** may adjust the residence time based on the reaction conditions such as, without limitation, feedstock, solvent ratio and temperature, and so on. In certain implementations, optimization of the residence time enables a complete depolymerization of the biomass without allowing further reactions to occur.

The temperature of the reaction within HTL converter **308** determines the depolymerization of the biomass to bio-oil, along with the repolymerization of the biomass into char. While the ideal reaction temperature is dependent on the feedstock used, temperatures above ideal can lead to an increase in char formation and eventually increased gas formation, while lower than ideal temperatures reduce depolymerization and overall product yields. In various implementations, controller **344** automatically determines the optimal conditions for producing an optimal biofuel quality and/or yield.

Controller **344** may also control the rate of heating of the biomass in the HTL converter **308** to potentially control the effects of secondary reactions that may become dominant where heating rates that are too low, which may lead to the formation of char. While high heating rates may be required to form liquid bio oil, controller **344** may set a threshold heating rate and temperature where liquid production is inhibited while gas production is favored in secondary reactions. Controller **344** may regulate pressure with temperature to determine the super or subcritical state of solvents, the overall reaction kinetics, and/or the energy inputs required to produce the desired processed biomass, e.g., biofuel.

While water acts as a catalyst in the HTL reaction, other catalysts can be added to the HTL converter **308** and/or conduit **310** to optimize the biomass to biofuel conversion. System **300** may include catalysts having water-soluble inorganic compounds and salts, including KOH and Na₂CO₃, and/or transition metal catalysts using Ni, Pd, Pt, and Ru supported on either carbon, silica or alumina within the HTL converter **308** and/or conduit **310**. The addition of these catalysts can increase bio oil yield by 20% or more due to, for example, the catalysts converting the protein, cellulose, and hemicellulose into oil.

Biofuels that are produced through hydrothermal liquefaction are carbon neutral in that there are no net carbon

emissions produced when burning the biofuel. The algae used to produce the bio oils use photosynthesis to grow and, therefore, consume carbon dioxide from the ocean and/or atmosphere. The burning of the biofuels produced from algae or other plants releases carbon dioxide into the atmosphere. However, this is almost completely offset by the carbon dioxide consumed from growing the algae or plants. Furthermore, hydrothermal liquefaction is a clean process that does not produce harmful compounds, such as ammonia, NO_x, or SO_x. Instead, the heteroatoms, such as nitrogen, sulfur, and chlorine, are converted into harmless byproducts including N₂ and inorganic acids that can be neutralized with bases.

FIG. 4 shows system **400** for converting a biomass into biofuel using a geothermal heat source **402**. System **400**, however, uses a second conduit **412** and heat exchanger **434** to capture thermal energy from geothermal heat source **402**. Conduit **412** uses a superheated fluid to transfer the thermal energy to HTL converter **408** to convert the biomass into a processed biomass, e.g., biofuel. By using a second conduit **412** and heat exchanger **434**, system **400** is arranged to control the temperature within HTL converter **408** by controlling the temperature of the superheated fluid being transported to the HTL converter **408** within the second conduit **412**. Controller **450** controls the temperature and/or flow rate of the superheated fluid within conduit **412** by controlling the operations of flow regulator **438** and temperature regulator **440**. Flow regulator **438** may operate and have components similar to flow regulators **322** and **324**. Temperature regulator **440** may include a heat exchanger and component like heat exchanger **326**. Controller **450** may control the flow rate of cooling water through regulator **440** to control the temperature of the superheated fluid exiting regulator **440** that is then transported to HTL converter **408**.

System **400** includes a biomass processing station **404** arranged to receive the biomass from a biomass harvester **406** and then output the biomass to HTL converter **408**. The processing station **404** also receives a processed biomass from the HTL converter **408**. The system **400** uses a first conduit **410** to transport the biomass from the biomass processing station **404** to the HTL converter **408** and transport the processed biomass from the HTL converter **408** to the biomass processing station **404**. Biomass processing station **404** may output the processed biomass via outlet **452** to a storage facility and/or processing facility for further processing. HTL converter **408** includes a heat exchanger arranged to transfer thermal energy from the superheated fluid in conduit **412** to the biomass being transported through conduit **410** to convert the biomass into the processed biomass. The superheated fluid is heated by the geothermal heat source **402** via heat exchanger **434**. Controller **450** is arranged to monitor one or more conditions of the biomass via sensors **414**, **416**, **418**, **420**, and **422** at the one or more locations along the conduit **410** and adjust operations of one or more components **422**, **424**, and **426** along the conduit **410** to, thereby, adjust the one or more conditions of the biomass at the one or more locations along the conduit **410**.

The conditions monitored by sensors **414-422** may include, without limitation, temperature, rate of temperature change, pressure, flow rate, catalyst concentration, and/or residence time the biomass within HTL converter **408**. Controller **450** may monitor the one or more conditions of the biomass by receiving sensor data associated with one or more conditions of the biomass from one or more of sensors **414-422**. The biomass may include an algae while the processed biomass may include bio oil, bio crude, and/or

biofuel. The conversion of the biomass to biofuel may include depolymerization of the biomass to bio oil.

In various implementations, controller **450** is configured to adjust the operations of one or more of components **424**, **426**, and **428** to optimize conditions of the biomass and, thereby, optimize a yield of the processed biomass. Components **424** and **426** may include flow regulators such as, for example, a pump and/or a flow control valve. A pump may be a variable speed pump to adjust flow rate within conduit **410**. A flow control valve may be adjustable to adjust flow rate with conduit **410**. Heat exchanger **428** may be controlled by controller **450** to regulate an output temperature of from heat exchanger **428** before the processed biomass, e.g., biofuel, is returned to processing station **404**. Heat exchanger **428** may receive cooling water at a temperature set by the depth of water inlet **430** and pass water through to extract heat from the processed biomass also passing through heat exchanger **428** via conduit **410**. The flow rate of cooling water through heat exchanger **428** may be regulated by a flow regulator controlled by controller **450**. The heated cooling water is then expelled into the surround body of water **442**, e.g., the ocean via outlet **432**. In this way, the temperature of the processed biomass is reduced before handling at processing station **404**. Although not shown in FIG. **4**, controller **450** is in electrical communications, e.g., via electrical conduits, with sensors **414-422** and **436**, and components **322-328**, and **438-440** to enable electrical communications associated with controlling conditions of the biomass in system **400**.

In various implementations, controller **450** monitors one or more conditions of the superheated fluid at one or more locations along the second conduit **412** via sensor **436** and adjusts operations of one or more components **438** and **440** along the second conduit **412** adjust the one or more conditions of the superheated fluid at the one or more locations along the second conduit **410**. By adjusting the temperature of the superheated fluid in conduit **412**, the temperature within HTL converter **408** can be set as desired to facilitate more efficient HTL conversion of the biomass to biofuel in conduit **410**. One technical advantage of system **400** is that the temperature of the superheated fluid in the HTL converter **408** can be regulated in a more consistent manner regardless of possible fluctuations in thermal energy output from the geothermal heat source **402**. Controller **450** may utilize an AI application and/or deep learning techniques described with respect to systems **200** and **300** to automatically control and optimize biofuel production within system **400**.

As illustrated in FIG. **4**, HTL converter **408** may reside within a body of water **442** such as the ocean where a substantial portion of conduit **410** is surrounded by the body of water **442**. While FIG. **4** shows the harvester **406** and biomass processing station **404** on land **444**, the biomass processing station **404** and biomass harvester **406** may reside in proximity with each other within the body of water **444**. The biomass processing station **404** may be located on a vessel, rig, or platform on the body of water **442**. The harvester **406** may include one or more photobioreactors or deepwater photobioreactors in the body of water **442** and in proximity to the system **400** and/or station **404**. The harvester **406** may be less than or equal to about 5 miles (about 8 Km), 3 miles (about 4.8 Km), 1 mile (about 1.6 Km), 0.5 miles (about 0.8 Km), 0.25 miles (about 0.4 Km) from system **400** and/or biomass processing station **404**.

Although heat exchanger **434** is shown adjacent to geothermal heat source **402**, heat exchanger **434** may extend through source **402** within the earth **448** below the ocean

floor **446**. Heat exchanger **434** may include a section of conduit **412** that extends through and/or near source **402**. The location of heat exchanger **434** with respect to source **402** may depend on the desired amount of heat exchange and temperature needed to convert the biomass into a processed biomass. Heat exchanger **434** may include a single pipe (i.e., section of conduit **412**) that extends through a region of the geothermal heat source **402**. The pipe may be substantially straight, U-shaped, serpentine, and/or curved in portions. Heat exchanger **434** may include multiple parallel pipes and/or capillaries with a common inlet and outlet that interface with conduit **412**. Conduit **410** and/or conduit **412** may have one or more sections of pipe including at least one of Tungsten and Titanium.

In some implementations, system **400** may interface with geothermal sources emitting temperatures up to 900 C. Conduit **412** may use much larger diameter pipes than the traditional 200 mm used from oil drilling equipment. In one implementation, the superheated fluid includes a supercritical CO₂ (SCCO₂) fluid. Supercritical CO₂ has properties between a gas and a liquid. While it still flows like a liquid when under pressures of 100 atmospheres or more, the volumetric density is inversely proportional to temperature so it can act like a liquid but be one tenth the density. The lower temperatures of traditional and enhanced geothermal systems operate at around 200 C to 300 C, and the complexity of these traditional systems have reduced their practicality. In some implementations, system **400** uses a higher temperature fluid in conduit **412** including, for example, a SCCO₂ fluid, which provides for more efficient high temperature geothermal operations and biomass conversion. In some implementations, a volumetric density change of the SCCO₂ fluid means the flow rate in conduit **412** has to maintain mass flow, which at 50 C and 200 Bar is about 0.7844 g/cc and at 550 C is 0.1244 g/cc.

FIG. **5** shows a system **500** for converting a biomass into biofuel and for generating electrical power using a geothermal heat source **502**. In addition to including a biomass processing station **504**, system **500** includes a power turbine **556** and generator **558** configured to output electrical power via power outlet **560**. The power generated via generator **558** may be used to provide power to external loads and/or may be used to power one or more components of system **500**.

System **500** includes a biomass processing station **504** arranged to receive the biomass from a biomass harvester **506** and then output the biomass to HTL converter **508**. The processing station **504** also receives a processed biomass from the HTL converter **508**. The system **500** uses a first conduit **510** to transport the biomass from the biomass processing station **504** to the HTL converter **508** and transport the processed biomass from the HTL converter **508** to the biomass processing station **504**. Biomass processing station **504** may output the processed biomass via outlet **554** to a storage facility and/or processing facility for further processing. HTL converter **508** includes a heat exchanger arranged to transfer thermal energy from superheated fluid in conduit **512** to the biomass being transported through conduit **510** to convert the biomass into the processed biomass. The superheated fluid is heated by the geothermal heat source **502** via heat exchanger **534**. Controller **550** is arranged to monitor one or more conditions of the biomass via sensors **514**, **516**, **518**, **520**, and **522** at the one or more locations along the conduit **510** and adjust operations of one or more components **522**, **524**, and **526** along the conduit **510** to, thereby, adjust the one or more conditions of the biomass at the one or more locations along the conduit **510**.

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The conditions monitored by sensors **514-522** may include, without limitation, temperature, rate of temperature change, pressure, flow rate, catalyst concentration, and/or residence time the biomass within HTL converter **508**. Controller **550** may monitor the one or more conditions of the biomass by receiving sensor data associated with one or more conditions of the biomass from one or more of sensors **514-522**. The biomass may include an algae while the processed biomass may include bio oil, bio crude, and/or biofuel. The conversion of the biomass to biofuel may include depolymerization of the biomass to bio oil.

In various implementations, controller **550** is configured to adjust the operations of one or more of components **524**, **526**, and **528** to optimize conditions of the biomass and, thereby, optimize a yield of the processed biomass. Components **524** and **526** may include flow regulators such as, for example, a pump and/or a flow control valve. A pump may be a variable speed pump to adjust flow rate within conduit **510**. A flow control valve may be adjustable to adjust flow rate with conduit **510**. Heat exchanger **528** may be controlled by controller **550** to regulate an output temperature of from heat exchanger **528** before the processed biomass, e.g., biofuel, is returned to processing station **504**. Heat exchanger **528** may receive cooling water at a temperature set by the depth of water inlet **530** and pass water through to extract heat from the processed biomass also passing through heat exchanger **528** via conduit **510**. The flow rate of cooling water through heat exchanger **528** may be regulated by a flow regulator controlled by controller **550**. The heated cooling water is then expelled into the surround body of water **562**, e.g., the ocean via outlet **532**. In this way, the temperature of the processed biomass is reduced before handling at processing station **504**. Although not shown in FIG. 5, controller **550** is in electrical communications, e.g., via electrical conduits, with sensors **514-522**, **536**, and **542**, and components **522-528** and **538-540** to enable electrical communications associated with controlling conditions of the biomass in system **500**.

In various implementations, controller **550** monitors one or more conditions of the superheated fluid at one or more locations along the second conduit **512**, via sensors **536** and **542**, and adjusts operations of one or more components **538** and **540** along the second conduit **512** adjust the one or more conditions of the superheated fluid at the one or more locations along the second conduit **510**. By adjusting the temperature of the superheated fluid in conduit **512**, the temperature within HTL converter **508** can be set as desired to facilitate more efficient HTL conversion of the biomass to biofuel in conduit **510**. One technical advantage of system **500** is that the temperature of the superheated fluid in the HTL converter **508** can be regulated in a more consistent manner regardless of possible fluctuations in thermal energy output from the geothermal heat source **502**. Controller **550** may utilize AI applications and/or deep learning techniques described with respect to systems **200** and **300** to automatically control and optimize biofuel production within system **500**. The superheated fluid within conduit **512** may also be used to drive turbine **556** which, in turn, drives a rotor with generator **558** to produce electrical power.

As illustrated in FIG. 5, HTL converter **508** may reside within a body of water **562** such as the ocean where a substantial portion of conduit **510** is surrounded by the body of water **562**. While FIG. 5 shows the harvester **506** and biomass processing station **504** on land **546**, the biomass processing station **504** and biomass harvester **506** may reside in proximity with each other within the body of water **562**. The biomass processing station **504** may be located on

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a vessel, rig, or platform on the body of water **562**. The harvester **506** may include one or more photobioreactors or deepwater photobioreactors in the body of water **562** and in proximity to the system **500** and/or station **504**. The harvester **506** may be less than or equal to about 5 miles (about 8 Km), 3 miles (about 4.8 Km), 1 mile (about 1.6 Km), 0.5 miles (about 0.8 Km), 0.25 miles (about 0.4 Km) from system **500** and/or biomass processing station **504**.

Although heat exchanger **534** is shown adjacent to geothermal heat source **502**, heat exchanger **534** may extend through source **502** within the earth **552** below the ocean floor **548**. Heat exchanger **534** may include a section of conduit **512** that extends through and/or near source **502**. The location of heat exchanger **534** with respect to source **502** may depend on the desired amount of heat exchange and temperature needed to convert the biomass into a processed biomass. Heat exchanger **534** may include a single pipe (i.e., section of conduit **512**) that extends through a region of the geothermal heat source **502**. The pipe may be substantially straight, U-shaped, serpentine, and/or curved in portions. Heat exchanger **534** may include multiple parallel pipes and/or capillaries with a common inlet and outlet that interface with conduit **512**. Conduit **510** and/or conduit **512** may have one or more sections of pipe including at least one of Tungsten and Titanium.

In some implementations, system **500** may interface with geothermal sources emitting temperatures up to 900 C. Conduit **512** may use much larger diameter pipes than the traditional 200 mm used from oil drilling equipment. In one implementation, the superheated fluid includes a supercritical CO₂ (SCCO₂) fluid. Supercritical CO₂ has properties between a gas and a liquid. While it still flows like a liquid when under pressures of 100 atmospheres or more, the volumetric density is inversely proportional to temperature so it can act like a liquid but be one tenth the density. The lower temperatures of traditional and enhanced geothermal systems operate at around 200 C to 300 C, and the complexity of these traditional systems have reduced their practicality. In some implementations, system **500** uses a higher temperature fluid in conduit **512** including, for example, a SCCO₂ fluid, which provides for more efficient high temperature geothermal operations and biomass conversion. In some implementations, a volumetric density change of the SCCO₂ fluid means the flow rate in conduit **512** has to maintain mass flow, which at 50 C and 200 Bar is about 0.7844 g/cc and at 550 C is 0.1244 g/cc.

FIG. 6 shows a cut-out view of a land-based system **600** for converting a biomass into biofuel using a geothermal heat source **602**. System **600** includes a biomass processing station **604** arranged to receive the biomass from a biomass harvester **606** and then output the biomass to a HTL converter **608** within the earth **616**. The processing station **604** also receives a processed biomass from the HTL converter **608**. The system **600** uses a conduit **610** to transport the biomass from the biomass processing station **604** to the HTL converter **608** and transport the processed biomass from the HTL converter **608** to the biomass processing station **604**. HTL converter **608** may include a heat exchanger including, for example, a section of conduit **610**, that is arranged to transfer thermal energy from geothermal heat source **602** to the biomass being transported through conduit **610** to convert the biomass into a processed biomass. System **600** also includes a controller **612** that is arranged to monitor one or more conditions of the biomass via sensors at the one or more locations along the conduit **610** and adjust operations of one or more components along the conduit **610** to, thereby, adjust the one or more conditions of the biomass at

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the one or more locations along the conduit 610. The sensors and components of system 600 may be configured and/or operate in a similar manner as the sensors and components described with respect to systems 300, 400, and 500. In some implementations, some of the sensors and/or components may reside in the biomass processing station 604. Biomass processing station may output and store processed biomass, e.g., biofuel, in a storage tank and/or reservoir 614.

FIG. 7 shows a process 700 for converting a biomass into biofuel using a geothermal heat source, such as heat source 302, 402, and 502. Process 700 includes: receiving, at a biomass processing station, the biomass from a biomass harvester (Step 702); transporting the biomass from the biomass processing station to a hydrothermal HTL converter via a conduit where the HTL converter is located remotely from the biomass processing station and within proximity to a geothermal heat source (704); transferring thermal energy from the geothermal heat source to the biomass to convert the biomass into a processed biomass (706); monitoring one or more conditions of the biomass at one or more locations along the conduit (708); adjusting operations of one or more components along the conduit, in response to the monitored conditions, to adjust the one or more conditions of the biomass at the one or more locations along the conduit (710); and receiving, at the biomass processing station, the processed biomass from the HTL converter via the conduit (712).

Elements or steps of different implementations described may be combined to form other implementations not specifically set forth previously. Elements or steps may be left out of the systems or processes described previously without adversely affecting their operation or the operation of the system in general. Furthermore, various separate elements or steps may be combined into one or more individual elements or steps to perform the functions described in this specification.

Other implementations not specifically described in this specification are also within the scope of the following claims.

What is claimed is:

1. A system for converting a biomass into a biofuel comprising:

a biomass processing station arranged to receive the biomass from a biomass harvester, output the biomass to a hydrothermal liquefaction (HTL) converter, and receive a processed biomass from the HTL converter; a conduit arranged to transport the biomass from the biomass processing station to the HTL converter and transport the processed biomass from the HTL converter to the biomass processing station, wherein a portion of the conduit is located below the surface of an ocean; and

the HTL converter including a heat exchanger arranged to transfer thermal energy from a geothermal heat source to the biomass to convert the biomass into the processed biomass, wherein the geothermal heat source is located below the surface of the ocean.

2. The system of claim 1 comprising a controller arranged to monitor one or more conditions of the biomass at the one or more locations along the conduit and adjust operations of one or more components along the conduit to, thereby, adjust the one or more conditions of the biomass at the one or more locations along the conduit.

3. The system of claim 2 comprising one or more sensors located at one or more sections of the conduit, the controller monitoring the one or more conditions of the biomass by

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receiving sensor data associated with one or more conditions of the biomass from the one or more sensor.

4. The system of claim 1, wherein the biomass includes an algae and the processed biomass includes bio oil, and wherein converting the biomass includes depolymerization of the biomass to bio oil.

5. The system of claim 2, wherein the controller is arranged to adjust the operations of the one or more components to optimize the one or more conditions of the biomass to optimize a yield of the processed biomass.

6. The system of claim 2, wherein the one or more components include at least one of a pump and a flow control valve.

7. The system of claim 2, wherein the one or more conditions of the biomass include at least one of temperature, pressure, flow rate, catalyst concentration, and residence time within the HTL converter.

8. The system of claim 1, wherein the geothermal heat source includes at least one of a volcano, volcanic vent, a hydrothermal vents.

9. The system of claim 1, wherein the HTL converter resides within the ocean.

10. The system of claim 9, wherein a substantial portion of the conduit is surrounded by the ocean.

11. The system of claim 1, wherein biomass processing station and biomass harvester are in proximity with each other within the ocean.

12. A system for converting a biomass into a biofuel comprising:

a biomass processing station arranged to receive the biomass from a biomass harvester, output the biomass to a hydrothermal liquefaction (HTL) converter, and receive a processed biomass from the HTL converter; a first conduit arranged to transport the biomass from the biomass processing station to the HTL converter and transport the processed biomass from the HTL converter to the biomass processing station;

the HTL converter including a first heat exchanger arranged to transfer thermal energy from a superheated fluid to the biomass to convert the biomass into the processed biomass;

a second conduit arranged to transport the superheated fluid to the HTL converter from a second heat exchanger in proximity to a geothermal heat source; and

the second heat exchanger being arranged to transfer thermal energy from the geothermal heat source to the superheated fluid.

13. The system of claim 12, wherein the biomass includes an algae and the processed biomass includes bio oil, and wherein converting the biomass includes depolymerization of the biomass to bio oil.

14. The system of claim 12, wherein the geothermal heat source includes at least one of a volcano, volcanic vent, a hydrothermal vents.

15. The system of claim 12, wherein the HTL converter resides within the ocean.

16. The system of claim 15, wherein a substantial portion of the conduit is surrounded by the ocean.

17. The system of claim 12, wherein biomass processing station and biomass harvester are in proximity with each other within the ocean.

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